

Scenario Based Activity

1. Median filter :- Removes salt and pepper noise by replacing each pixel with the median of its neighbours. It preserves image details while eliminating impulse noise.
2. Gaussian filter :- Smooths gaussian noise using a weighted average of surrounding pixels. It reduces sensor noise while maintaining overall image quality.
3. Bilateral filter :- Removes random noise while preserving fine edges in medical images. It smooths similar pixels without blurring boundaries.
4. Median filter :- Eliminates black and white dots from scanned documents. It restores clean and sharp document edges.
5. Non-local means filter :- Eliminates black and white dots from scanned documents. Reduces heavy grain in low light images by comparing similar image patches. It preserves important details better than simple smoothing.
6. Histogram Equalization :- Enhances contrast in foggy road images by spreading pixel intensity values. This improves visibility of roads and objects.

7. Contrast stretching:- Expands the image intensity range to make underwater image clearer. It improves brightness and colour visibility.

8. CLAHE (Contrast Limited Adaptive Histogram equalization)

Enhances local contents contrast in chest X-rays without over-amplifying noise. It makes internal structure easier to observe.

9. Histogram Equalization:- Increases intensity variation in night-time traffic images. It makes vehicles and road features more visible.

10. Contrast sketching:- Expands the narrow grayscale intensity range across the full spectrum. This improves the overall and detail of the range image.